*A (More) Objective View of AI*

# Implications for the Future

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# 1. Introduction

# AI Is Not Going Away

- **AI is not going away**. Blindly refusing AI is not the solution. Instead, it is fundamental to understand its capabilities and limitations.

> *"Love it or hate it, artificial intelligence (AI) is here, and it's not going away. As the technology evolves, AI will only become more prominent in our everyday interactions, shaping everything from how students learn to the work employees do at the office."* [1]
>
> — **AI Won't Take Your Job if You Know About IA**, Harvard Graduate School of Education, 2024

[1] `AI Won't Take Your Job if You Know About IA` - Dede & McCool, Harvard Graduate School of Education, 2024

# 2. Unlikely Exponential Growth

- **Fundamental Physical Limits of Transistors and Memory**.
  - The thermodynamic minimum gate length of a *transistor* will likely be reached by 2030, while the limits of photolithography are expected around 2029. [2]
  - *Memory bandwidth* has been increasing far more slowly than FLOP/s, at 30x in the last 20 years (1.5x/2yr), while FLOP/s have increased 90,000x in the same period (3x/2yr). This is a much more conservative constraint on potential AI growth. [3, 4]

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[2] The Transistor Cliff - S. Constantin, 2023

[3] AI and Memory Wall - Gholami et al., IEEE Micro, 2024

[4] Challenges and Research Directions for Large Language Model Inference Hardware - Ma and Patterson, Google, 2026

- **Hardware Resource Constraints**. Reconstructing one cubic millimeter of human temporal cortex (~50-57k cells and ~130-150M synapses) generated 1.4 Petabytes of data. Scaling to the full human brain would require 1.6 zettabytes of storage costing $50 billion and spanning 140 acres. [5, 6]

- **Datacenters require 24/7 energy**. Solar and wind energy are not suitable for this purpose. On average, a nuclear power plant takes about 9.4 years to build, with an additional 3-4 years for the licensing process. [7]

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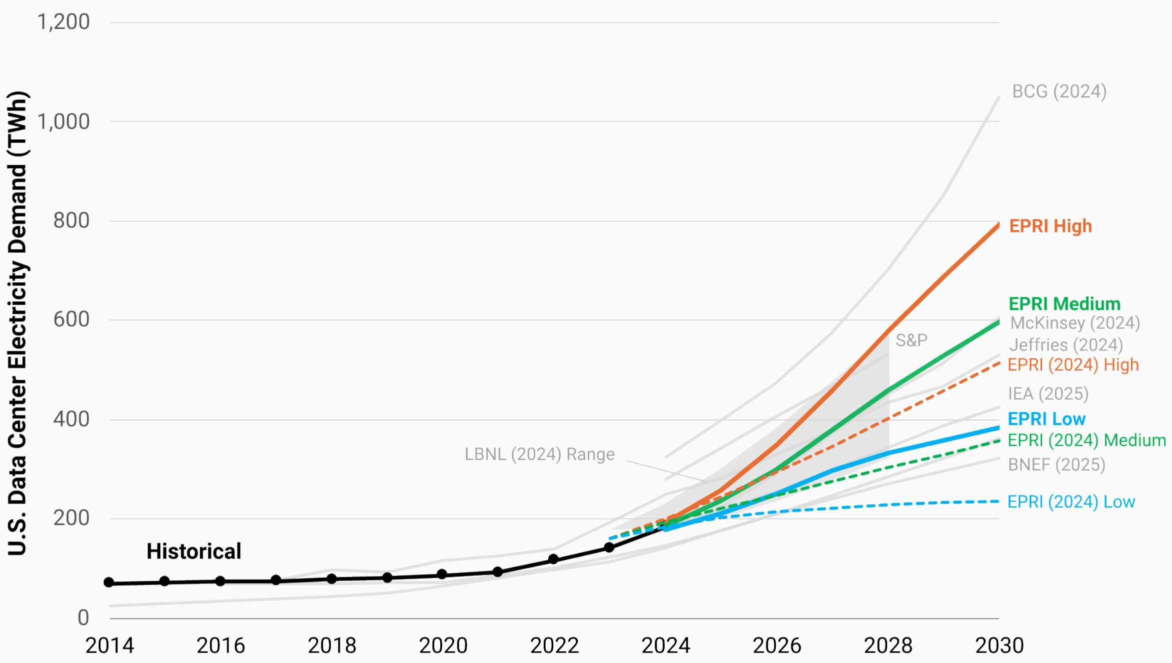
[5] A Browsable Petascale Reconstruction of the Human Cortex - Blakely et al., Google, 2021

[6] Full scan of 1 cubic millimeter of brain tissue took 1.4 petabytes of data, equivalent to 14,000 4K movies - S. Grimm, Tom's Hardware, 2024

[7] Global nuclear reactor construction starts and duration, 1949-2023 - Cleveland & Mirkova, Boston University, 2024

"***The data center share of U.S. total electricity demand in 2030 ranges from 9% to 17%, an increase from 4-5% today***. *At the state level, continued development of the largest DC market in Virginia implies a share increasing to between 39% and 57% by 2030.*"




Powering Intelligence 2026 - EPRI, 2026

- **Running out of data**. AI training will exhaust text data in a short timeframe.

> *"if rapid growth in dataset sizes continues,* ***models will utilize the full supply of public human text data at some point between 2026 and 2032****, or one or two years earlier if frontier models are overtrained. At this point, the availability of public human text data may become a limiting factor in further scaling of language models."* [9]
>
> — **The Projected Impact of Generative AI on Future Productivity Growth**, University of Pennsylvania, 2025

[9] Will we run out of data? Limits of LLM scaling based on human-generated data - Villalobos et al., EpochAI, 2024

- **AI quality degradation when trained on recursively generated data**. AI-generated content has surpassed the *quantity* of human-written articles. This implies that AI models are trained on increasingly low-quality data, reinforcing hallucinations. [10, 11, 12, 13]

> *"The 35% prevalence of AI-generated and AI-assisted text transforms the theoretical risk of model collapse, wherein future AI models degrade after recursively ingesting AI-generated data, into an empirical concern"* [14]
>
> — **The Impact of AI-Generated Text on the Internet**, Dolezal et al., 2026

[10] More Articles Are Now Created by AI Than Humans - Paredes et al., Graphite, 2025

[11] AI models collapse when trained on recursively generated data - Shumailov et al., Nature, 2024

[12] A Shocking Amount of the Web is Machine Translated: Insights from Multi-Way Parallelism - Thompson et al., Association for Computational Linguistics, 2024

[13] 74% of New Webpages Include AI Content (Study of 900k Pages) - Law et al., Ahrefs, 2025

[14] The Impact of AI-Generated Text on the Internet - Dolezal et al., 2026

# 3. Productivity Gain Uncertainty

# Productivity Gain Uncertainty

- The relationship between AI adoption and evidence of productivity gains is a *controversial topic*. Some research studies and economic data suggest that AI enhances productivity, while others show no benefits or even a negative impact.

> *"The AI productivity story in early 2026 is neither the revolution that vendors promise nor the disappointment that skeptics predict **-it's a transition whose timeline remains genuinely uncertain**.*
>
> *… History suggests AI's macro impact may similarly require a decade of complementary innovation before the statistics catch up to the promise."* [15]
>
> — **AI Productivity's $4 Trillion Question: Hype, Hope, And Hard Data**, Forbes, 2026

[15] AI Productivity's $4 Trillion Question: Hype, Hope, And Hard Data - G. Yildiz, Forbes, 2026

> *"Combine the increase in working hours spent using generative AI with how much it improves efficiency, and you get a boost of about* ***0.25-0.5 percentage points to productivity growth over the past year. This calculation is almost certainly too generous.****"* [16]
>
> — **The AI productivity boom is not here (yet)**, The Economist, 2026

[16] The AI productivity boom is not here (yet) - The Economist, 2026

*"AI will increase productivity and GDP by **1.5% by 2035**, nearly 3% by 2055, and 3.7% by 2075. AI's boost to annual productivity growth is strongest in the early 2030s but eventually fades"* [17]

— **The Projected Impact of Generative AI on Future Productivity Growth**, University of Pennsylvania, 2025

*"Surprisingly, we find that when developers use AI tools, they **take 19% longer than without – AI makes them slower**."* [18]

— **Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity**, METR, 2025

[17] The Projected Impact of Generative AI on Future Productivity Growth - Arnon et al., University of Pennsylvania, 2025

[18] Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity - Becker et al., METR, 2025

> *"Despite widespread experimentation,* ***only one-in-eight (12%) CEOs say AI has delivered both cost and revenue benefits****. Overall, 33% report gains in either cost or revenue, while 56% say they have seen no significant financial benefit to date."* [19]
>
> — **PwC 2026 Global CEO Survey**, (4,454 executives in 95 countries)

[19] CEO confidence in revenue outlook hits five-year low - PwC 2026 Global CEO Survey, 2026

*"Firms report little impact of AI over the last 3 years, with **over 80% of firms reporting no impact on either employment or productivity**. Firms predict sizable impacts over the next 3 years, forecasting AI will boost productivity by 1.4%"* [20]

— **Firm Data on AI**,
National Bureau of Economic Research, 2026
(6000 senior executives, across US, UK, Germany, Australia)

[20] Firm Data on AI - National Bureau of Economic Research, 2026

*"AI tools didn't reduce work, they **consistently intensified it**.*

*... We identified three main forms of intensification: task expansion, blurred boundaries between work and non-work, and more multitasking.*

*... For workers, the cumulative effect is fatigue, burnout, and a growing sense that work is harder to step away from."* [21]

— **AI Doesn't Reduce Work-It Intensifies It**, Harvard Business Review

[21] AI Doesn't Reduce Work-It Intensifies It - Ranganathan & Ye, Harvard Business Review, 2026

***Work is accelerating faster than the systems designed to manage it.***

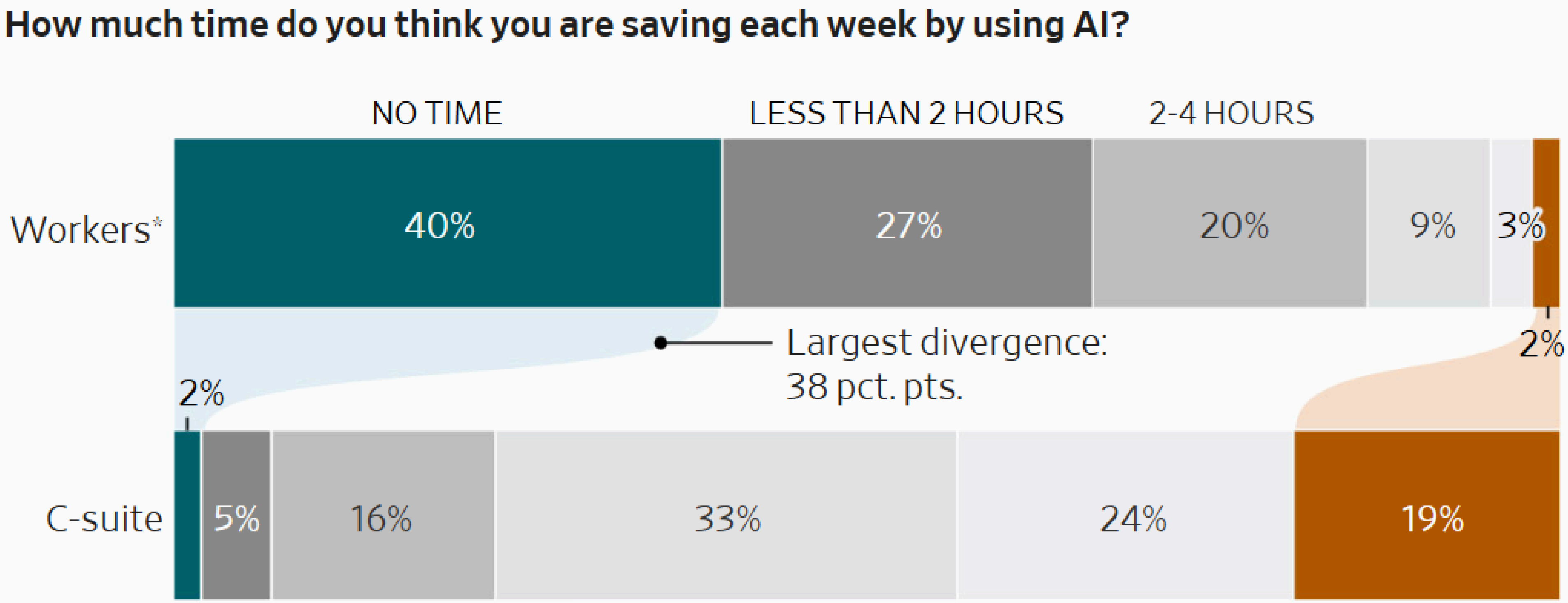
*... Focus efficiency, the percentage of work time spent in focused, uninterrupted activity, declined to 60% – a three-year low. Risk of disengagement jumped 23%.*

*... AI is adding to workloads rather than redistributing them. Collaboration is expanding faster than attention can support it. Productivity gains are real but increasingly funded by fragmentation rather than depth.*

*... The data is unambiguous: AI does not reduce workloads. AI is being used as an additional productivity layer, not a substitute for existing work.* [22]

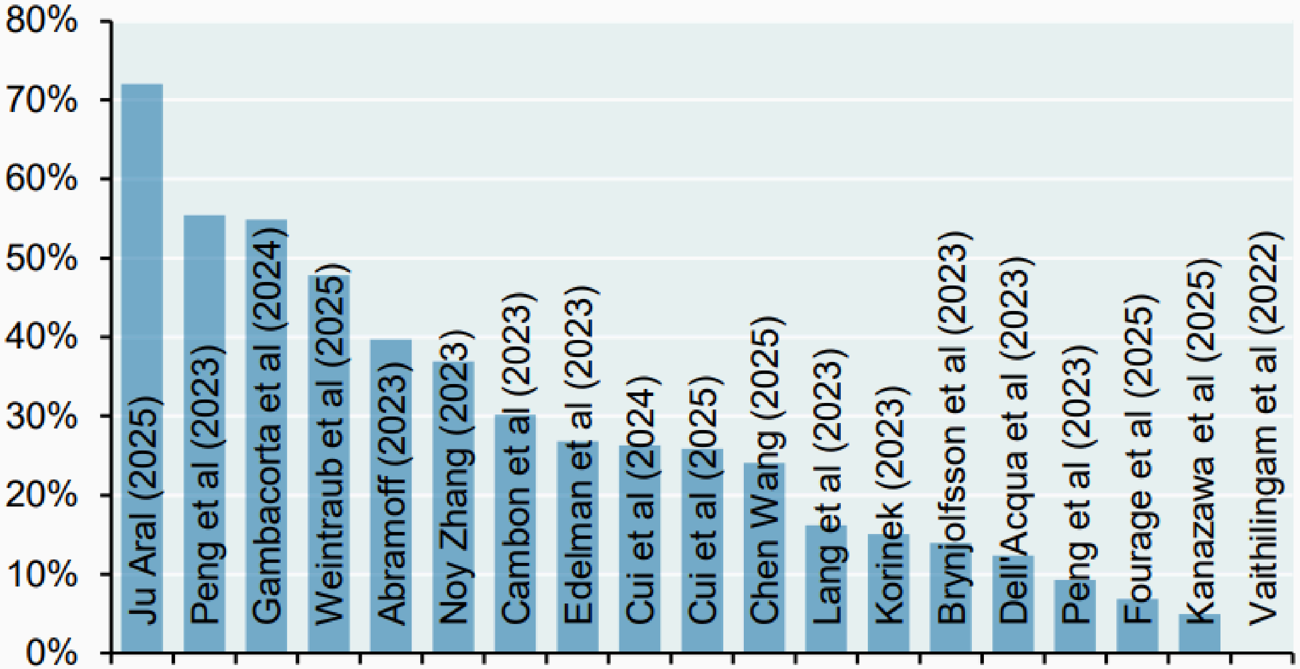
— **2026 State of the Workplace**, ActivTrak, 2026

[22] 2026 State of the Workplace - ActivTrak, 2026



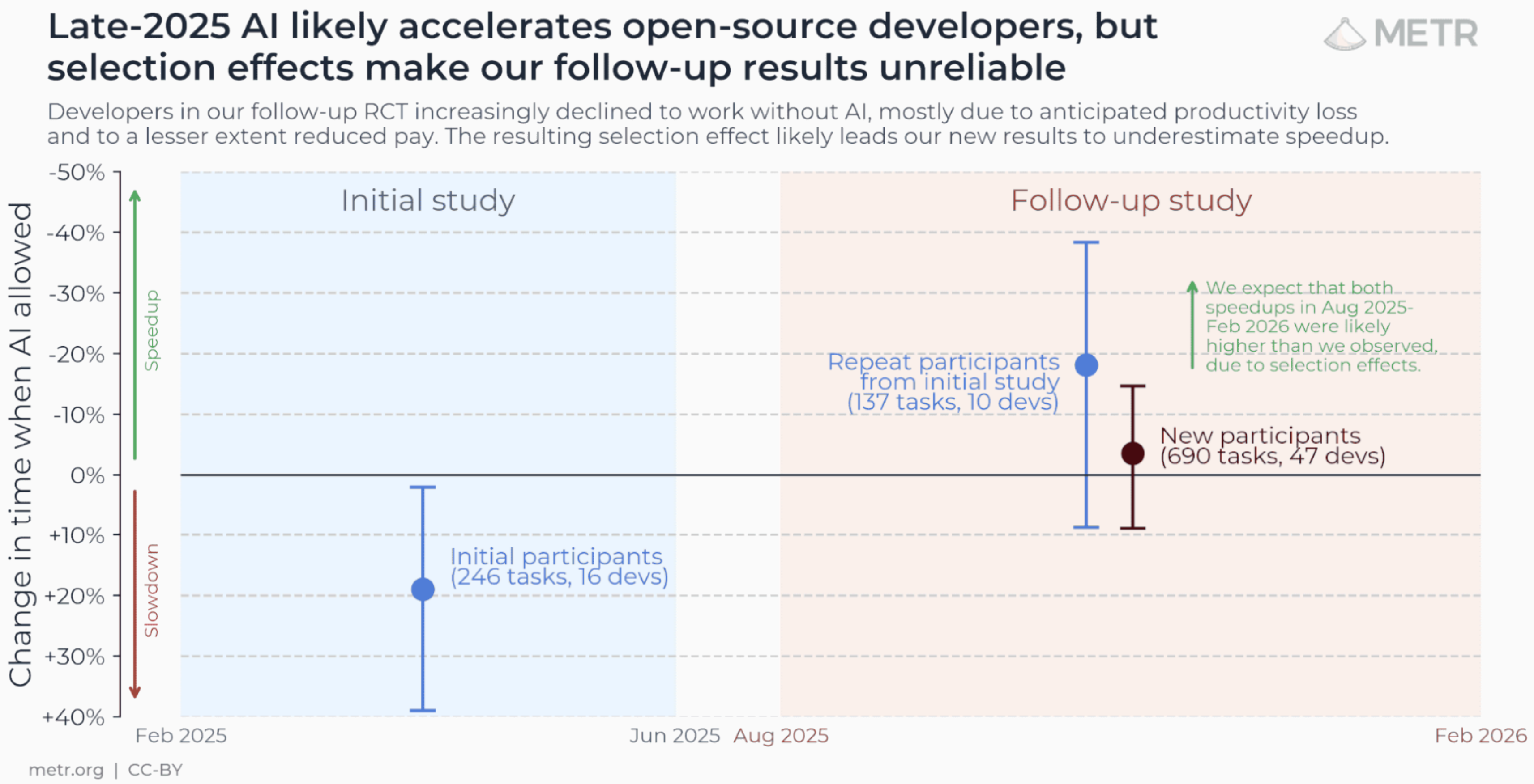

CEOs Say AI Is Making Work More Efficient. Employees Tell a Different Story - L. Ellis, The Wall Street Journal, 2026

Academic estimates of the effect of AI on labor productivity, Percent




Source: GS Global Economics Research, December 2025

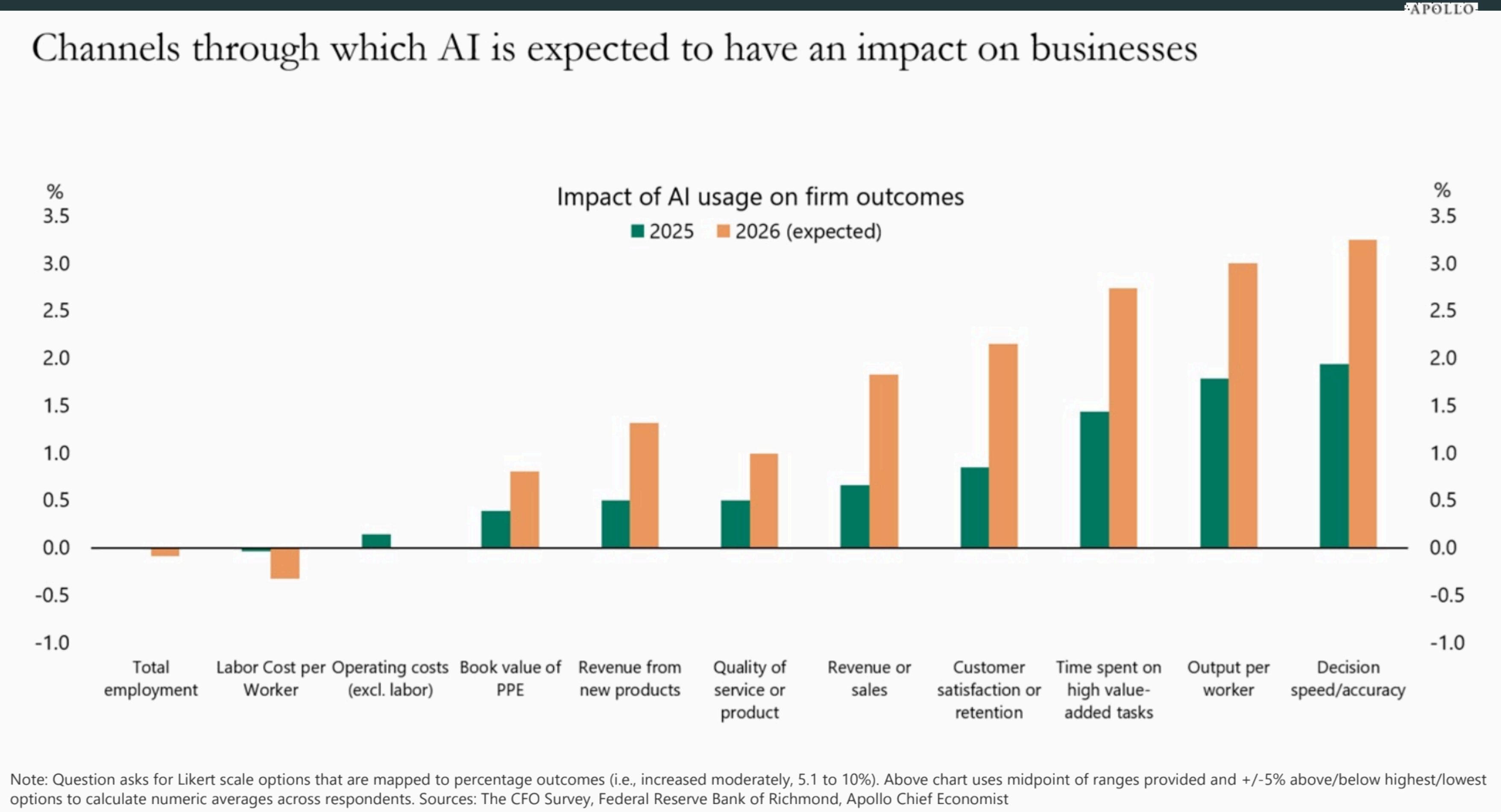
Smothering Heights - M. Cembalest, J.P. Morgan, 2026




We are Changing our Developer Productivity Experiment Design - Becker et al., METR, 2026

> *Across 515 high-growth startups, we run a field experiment in which treated firms receive information about how other firms have reorganized production around AI, prompting them to search for use cases across a broader set of firm functions. We find that treated firms discover more AI use cases, a 44% increase, concentrated in product development and strategy. These changes result in economically meaningful performance gains.* ***Treated firms complete 12% more tasks, are 18% more likely to acquire paying customers, and generate 1.9x higher revenue.*** [26]
>
> — **Mapping AI into Production: A Field Experiment on Firm Performance**, The Business School of the World, 2026

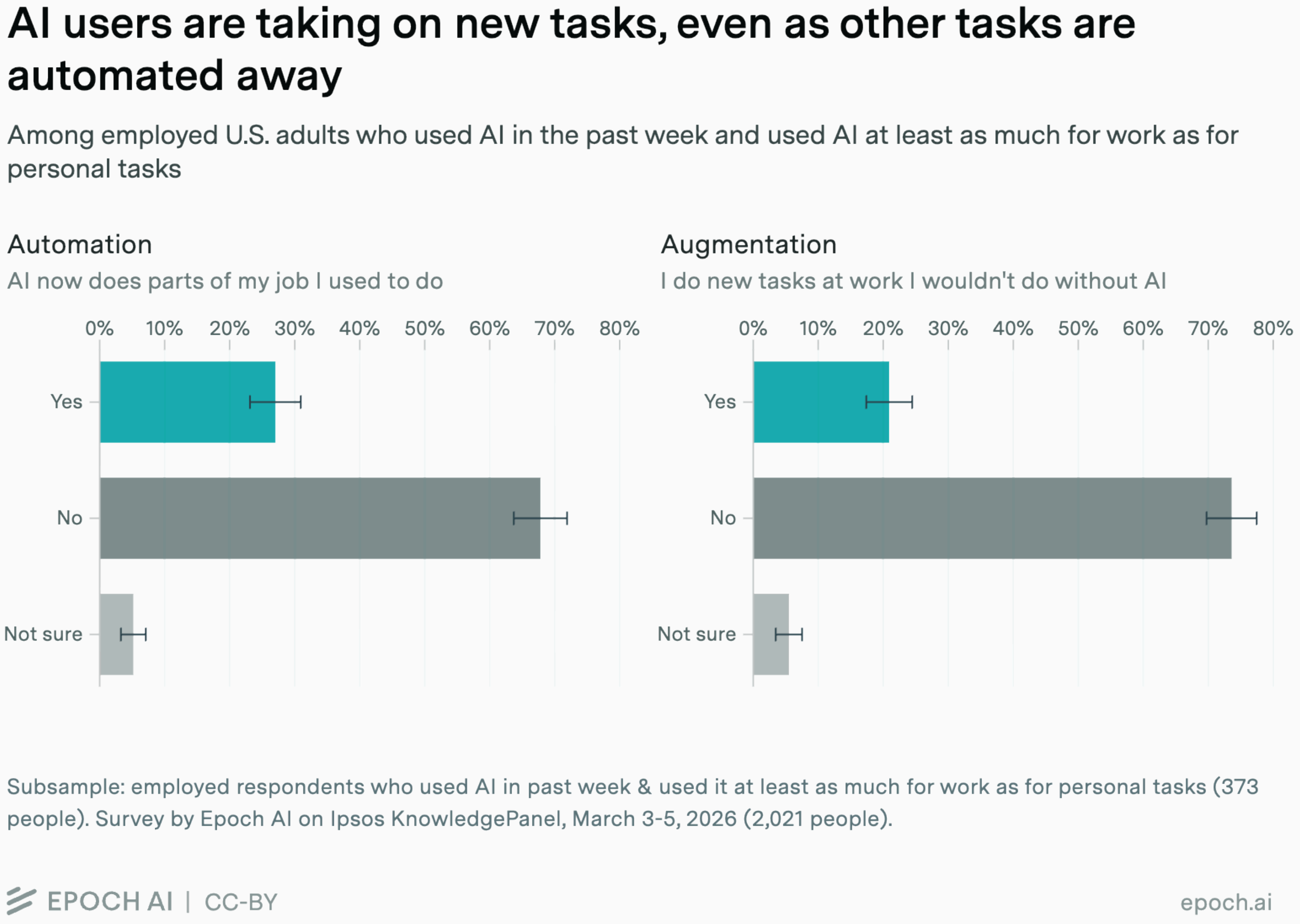
[26] Mapping AI into Production: A Field Experiment on Firm Performance - Kim et al., The Business School of the World, 2026




The AI Revolution Continues - T. Slok, Apollo Global Management, 2026

# Productivity Gain Uncertainty - Increased by AI






Half of employed AI users now use it for work - E. & C. Falkman Olsson, Epoch AI, 2026

# 4. Employment

# Productivity Gain and Labor Markets

*"Across domains,* ***AI is better at amplifying human judgment than at replacing it****, rewarding those who can evaluate outputs and decide how to direct them toward useful ends.*

*... By making execution cheap, AI shifts value upstream toward cognitive tasks-judgment, problem framing, and integration–that are unevenly distributed and slow to acquire. As a result, access to AI alone is unlikely to equalize outcomes.* ***What matters is not whether workers can use AI but whether they can turn its output into useful work.****"*

— **AI raises the productivity bar**,
Science, 2026

AI raises the productivity bar - Wu & Vasilescu, Science, 2026

*“AI has become the most powerful proactive frame available. ‘We’re restructuring around AI’ is a growth signal. ‘We over-hired during the pandemic and revenue softened’ is an accountability signal.*

*… What makes AI washing corrosive is the confusion it creates, both inside and outside companies.*

*… **And every incoherent account adds to the public’s conviction that AI is eliminating jobs at a pace the data simply do not support.**“* [30]

— **The 'AI-Washing' of Job Cuts Is Corrosive and Confusing**, Bloomberg, 2026

[30] The 'AI-Washing' of Job Cuts Is Corrosive and Confusing - G. Mukunda, Bloomberg, 2026

> “***A flood of sometimes conflicting analyses shows the yawning gap between what little is known about how AI is changing work and everyone’s understandable hunger for certainty***.
>
> *… Economists say it’s nearly impossible to forecast AI’s effect on the labor market from the current capabilities of the technology or the business sectors it’s seeping into first.*“ [31]
>
> — **See which jobs are most threatened by AI and who may be able to adapt**, The Washington Post, 2026

[31] See which jobs are most threatened by AI and who may be able to adapt - Schaul & Shira, The Washington Post, 2026

*“**we’re sceptical that firms can quickly and seamlessly substitute workers with AI even in sectors where the potential for AI disruption is greatest**. What’s more, some surveys suggest that AI use in larger US firms has recently stalled.*

*… Data from Challenger, Gray and Christmas suggests that AI-related US job losses are snowballing. In the first 11 months of 2025, AI was cited as a reason for almost 55,000 US job cuts, which accounts for over 75% of the reported AI-related job losses since the reason was first cited in 2023. However, these 55,000 or **so many AI-related job losses above account for only 4.5% of total reported job losses in the report**.“* [32]

— **Evidence of an AI-driven shakeup of job markets is patchy**, Oxford Economics, 2026

[32] Evidence of an AI-driven shakeup of job markets is patchy - Oxford Economics, 2026

> *"Despite headlines,* ***AI isn't the culprit behind slow hiring****. LinkedIn data shows* ***economic uncertainty, and monetary policy shifts are the primary drivers****. Advanced economies are struggling the most, with hiring down 20%-35% compared to pre-pandemic levels.*
>
> *... Many roles as we have known them will undergo this transformation into new collar. In the past two years,* ***employers have created at least 1.3 million AI-related job opportunities.****"* [33]
>
> — **Labor Market Report**, LinkedIn, 2026

[33] Labor Market Report - LinkedIn, 2026

*We find that thus far, **there is no evidence of a reduction in job postings for industries or firms which have higher levels of AI adoption**. The overall slowdown in national job postings following the pandemic recovery does not appear to be driven (even modestly) by AI.* [34]

— **FEDS Notes**, 2026

[34] FEDS Notes - Board of Governors of the Federal Reserve System, 2026

*"Micro-level evidence suggests that AI generates* ***meaningful time savings*** *for many workers. At the macro level, in recent years industries with higher AI adoption rates have* ***experienced faster productivity growth****. While we do not establish causality, this relationship is statistically significant and similar in magnitude in Europe and the US."* [35]

— **Mind the Gap: AI Adoption in Europe and the U.S.**, National Bureau of Economic Research, 2026

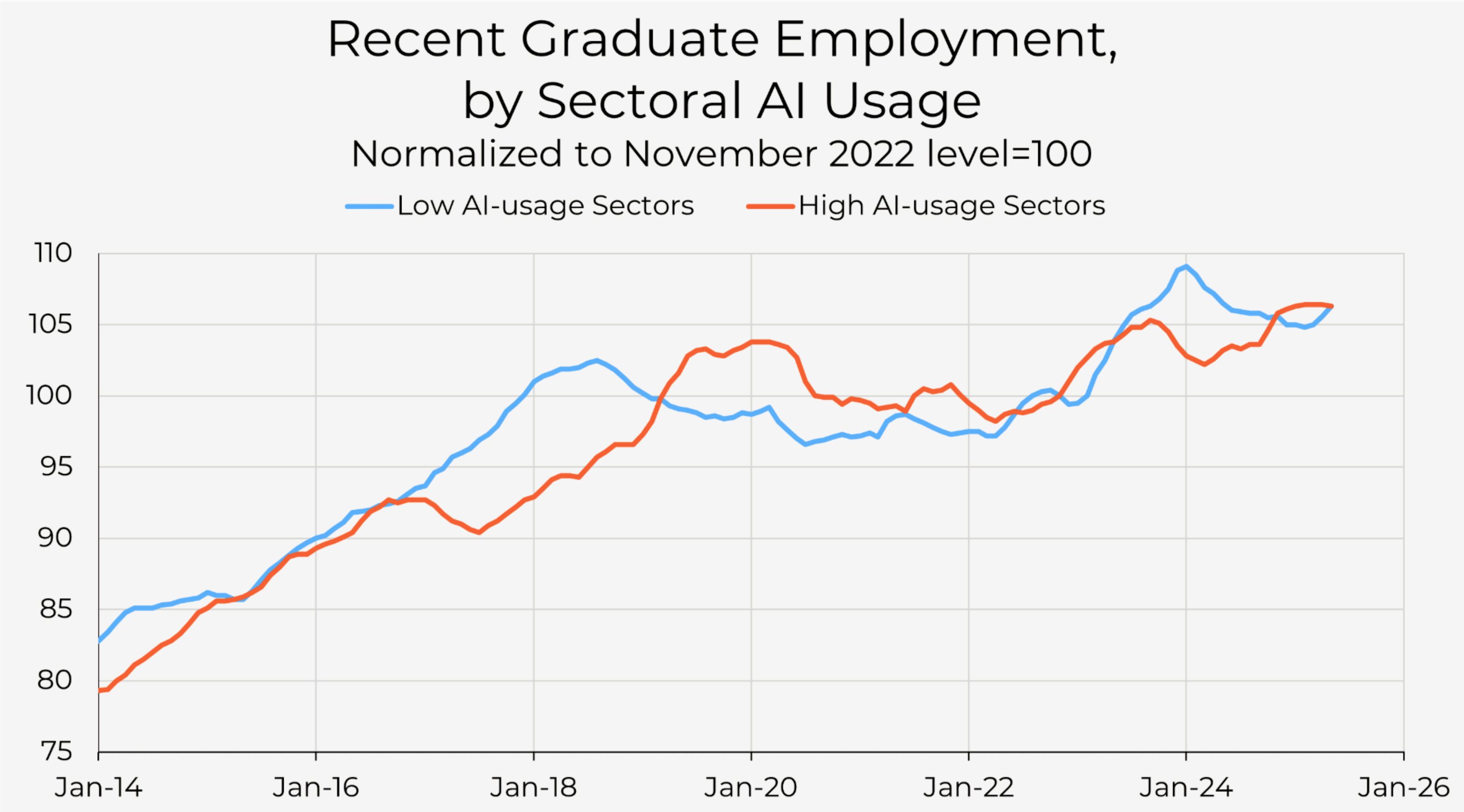
[35] Mind the Gap: AI Adoption in Europe and the U.S. - National Bureau of Economic Research, 2026

> *"While anxiety over the effects of AI on today's labor market is widespread, our data suggests it remains largely speculative. The picture of AI's impact on the labor market that emerges from our data is one that* ***largely reflects stability, not major disruption at an economy-wide level****."* [36]
>
> — **Evaluating the Impact of AI on the Labor Market: Current State of Affairs**, Budgetlab, Yale University, 2025

[36] Evaluating the Impact of AI on the Labor Market: Current State of Affairs - Gimbel et al., Budgetlab, Yale University, 2025

> *"Young workers of all education levels are lagging the rest of the labor market.* ***Focusing too much on education rather than age as the main labor market weakness starts us in the wrong direction****.*
>
> *So is AI nonetheless to blame for the broad-based weakness in the labor market for young people? It's true that some lower-skilled jobs can be replaced by AI. Call center workers and data entry jobs are potential examples. But there are not enough of these jobs to really drive the youth labor market."* [37]
>
> — **AI and Young-adult Jobs: The Real Mystery**, Economic Innovation Group, 2026

[37] AI and Young-adult Jobs: The Real Mystery - Ozimek & Goldschlag, Economic Innovation Group, 2026




Don't Blame AI For The Rise in Recent Graduate Unemployment - W. Raderman, Employ America, 2026 *"the association between LLM performance and task duration is well approximated by a relatively flat, near-linear relationship rather than a steep, wave-like pattern.*

*... the success rates achieved by LLMs in this analysis should not be interpreted as implying that a corresponding share of tasks can (or should) be automated today*

*...* ***the loss of individual tasks does not necessarily hurt the employees****. Indeed, based on the expertise of task and how that relates to the occupation's bundle of tasks, automation could increase or decrease wages.*

*... In particular,* ***we require each task instance to be self-contained****, with all relevant information provided in the prompt. This constraint limits our ability to represent tasks that depend on interaction with external artifacts"*

— **Crashing Waves vs. Rising Tides: Preliminary Findings on AI Automation from Thousands of Worker Evaluations of Labor Market Tasks**, Economic Innovation Group, 2026

Crashing Waves vs. Rising Tides: Preliminary Findings on AI Automation from Thousands of Worker Evaluations of Labor Market Tasks - Mertens et al., arXiv, 2026


Employment by industry vs prepandemic trend
Vertical line denotes launch of ChatGPT. Y axes vary.
Residential building construction
Veneer, plywood, and engineered wood product mfg
Household appliance manufacturing
Motor vehicle body and trailer manufacturing
Plastics product manufacturing
Furniture and home furnishing merchant wholesalers
Other motor vehicle dealers
Beer, wine, and liquor retailers
General freight trucking
Support activities for road transportation
Insurance carriers
Specialized design services
Computer systems design and related services
Employment services
Other support services
Drinking places, alcoholic beverages
2015
2020
2025
12-month trailing average. Trend estimated using data from 2012-2019. | Source: Bureau of Labor Statistics

Employment by Industry - B. Casselman, Chief Economics Correspondent for The New York Times, 2026

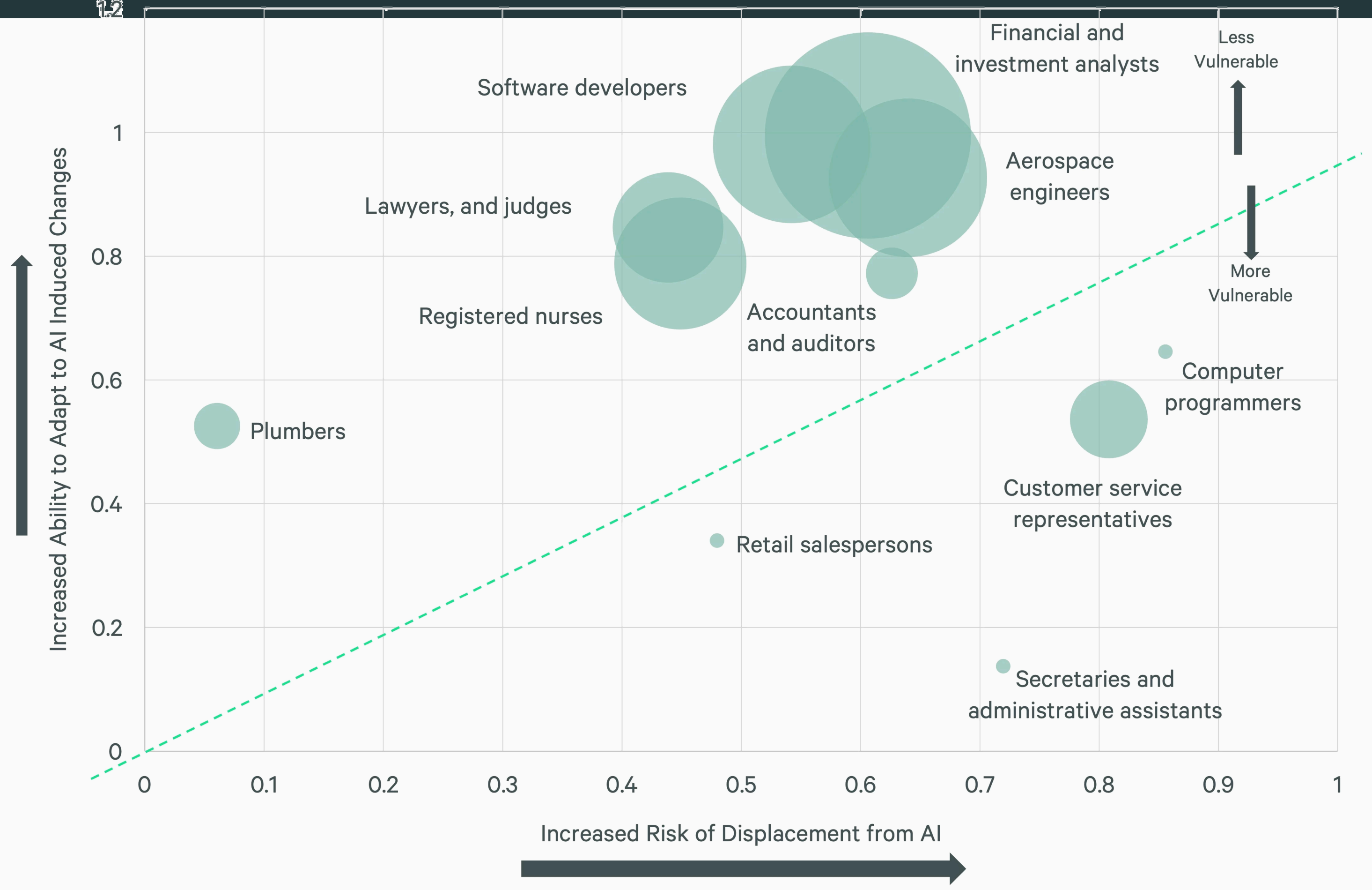
- *"Labor impacts are dominated by augmentation rather than substitution. Among firms shifting task structures, 66% engage exclusively in augmentation.*
- *Functional integration and operational investments are positively associated with sales increases, performance, and headcount declines. Conversely, worker-task breadth is positively associated with enhanced performance/sales, but not with displacing labor."*

— **The Microstructure of AI Diffusion**,
U.S. Census Bureau's Business Trends and Outlook Survey (BTOS),
1.2 million businesses firms, 2026

The Microstructure of AI Diffusion: Evidence from Firms, Business Functions, and Worker Tasks - Boney et al., U.S. Census Bureau, 2026

> *"Job postings are drawn from the Lightcast (formerly Burning Glass) database, which catalogues job postings from more than 65,000 sources*
>
> *... Despite the recent boom in AI investment across the economy and fears that the technology will lead to widespread job losses,* ***we find no evidence of negative impacts thus far on firms' job-posting behavior****."*
>
> — **FEDS Notes**, Board of Governors of the Federal Reserve System, 2026

FEDS Notes - Board of Governors of the Federal Reserve System, March, 2026




AI's Impact on the Economy, Employment & Productivity - CBRE, 2026

*"AI investment is rewriting the economic playbook in real time. Hyperscalers are projected to spend **$3.7 trillion on AI infrastructure over the next five years** — a buildout that experts estimate **will eclipse the railroad expansion of the 1850s** in relative scale. 2025 AI investment was equivalent to roughly half of U.S. GDP growth for the year."*

— **AI's Impact on the Economy, Employment & Productivity**, CBRE, 2026

# 5. Positive Impacts

> *"In 2020, Demis Hassabis and John Jumper presented an AI model called* ***AlphaFold2****. With its help, they have been able to* ***predict the structure of virtually all the 200 million proteins*** *that researchers have identified. Since their breakthrough,* ***AlphaFold2 has been used by more than two million people from 190 countries****. Among a myriad of scientific applications, researchers can now better understand antibiotic resistance and create images of enzymes that can decompose plastic."* [44]
>
> — **AlphaFold2**, Nobel Prize in Chemistry, 2024

[44] AlphaFold2 - Nobel Prize in Chemistry, 2024

# Material Discovery

*"With GNoME, we've multiplied the number of technologically viable materials known to humanity. Of its 2.2 million predictions, 380,000 are the most stable,* ***making them promising candidates for experimental synthesis****."* [45]

— **GNoME**, DeepMind, 2023

*"The open-access materials database … enabling AI-ready scientific datasets at an unprecedented scale for batteries, quantum computing, microelectronics, and more.*

*… in 650,000+ registered users, 32,000+ scientific journal citations, ~5,000 active users a day, 200,000+ materials, 577,000+ molecules, 465+ TBs of data, 300 million+ data requests a year."* [46]

— **The Materials Project**, Berkeley Lab, 2026

[45] Millions of new materials discovered with deep learning - Merchant & Cubuk, DeepMind, 2023

[46] Accelerating Discovery: How the Materials Project Is Helping to Usher in the AI Revolution for Materials Science - Merchant & Cubuk, DeepMind, 2023

> *"GraphCast, a state-of-the-art AI model able to make medium-range weather forecasts with unprecedented accuracy. GraphCast predicts weather conditions up to 10 days in advance* ***more accurately and much faster than the industry gold-standard weather simulation system****."* [47]
>
> — **GraphCast**, Google, 2023

[47] GraphCast: AI model for faster and more accurate global weather forecasting - Google, 2023

*"FourCastNet 3 delivers forecasting accuracy that surpasses leading conventional ensemble models and rivals the best diffusion-based methods, while* ***producing forecasts 8 to 60 times faster*** *than these approaches.*

*… a strong candidate for improving meteorological forecasting and early warning systems."* [48]

— **FourCastNet**,
NVIDIA, 2025

[48] FourCastNet 3: A geometric approach to probabilistic machine-learning weather forecasting at scale - BONEV ET AL., 2025

*"Idiopathic pulmonary fibrosis (IPF) is a type of chronic scarring lung disease characterized by a progressive and irreversible decline in lung function affecting around 5 million people globally.* [49]

*... This is the first time an **AI-designed drug** for an AI-discovered disease-associated target **has been tested in the clinic** [phase 2a]."* [50]

— **TNIK Inhibitor**,
INSILICO, 2025

[49] TNIK Inhibitor: Treating Fibrotic diseases (Phase IIa completed) - INSILICO, 2025

[50] A Phase 2 Readout Generates Excitement for the Potential of AI-Driven Drug Discovery - INSILICO, 2025

*"REDMOD [Radiomics-based Early Detection MODel] detects the subvisual signature of pre-clinical PDA [pancreatic ductal adenocarcinoma] a median of* ***475 days before clinical diagnosis****.*

*… The performance of* ***REDMOD surpasses that of radiologists, demonstrating nearly double the sensitivity*** *for detecting visually occult PDA, with an advantage that increases to* ***nearly threefold for cases detected more than 24 months prior to diagnosis****."* [51]

— **Next-generation AI for visually occult pancreatic cancer detection in a low-prevalence setting with longitudinal stability and multi-institutional generalisability**, BMJ Gut, 2026

[51] Next-generation AI for visually occult pancreatic cancer detection in a low-prevalence setting with longitudinal stability and multi-institutional generalisability - Mukherjee et al., BMJ Gut, 2026

> *"Using advanced statistical learning methods and a process known as 'joint learning,' the researchers' AI model was able to identify a specific set of proteins that form a general pattern for diseases involving brain degeneration.*
>
> *Vogel confirms that their* ***AI model outperforms previous models, while also being able to diagnose five different dementia-related conditions****: Alzheimer's disease, Parkinson's disease, ALS, frontotemporal dementia, and previous stroke."* [52]
>
> — **New AI model can detect multiple cognitive brain diseases from a single blood sample**, Lund University, 2026

[52] New AI model can detect multiple cognitive brain diseases from a single blood sample - Lund University, 2026

*"The AI tool was trained and validated in 72,000 patients from nine NHS trusts in England, who were followed up for a decade after their CT scans.* ***It predicted their risk of developing heart failure in the next five years with 86% accuracy.*** *"* [53, 54]

— **Early Prediction of Heart Failure From Routine Cardiac CT Using Radiomic Phenotyping of Epicardial Fat**,
OIKONOMOU ET AL., JOURNAL OF THE AMERICAN COLLEGE OF CARDIOLOGY, 2026

[53] Scientists develop AI tool to spot heart failure risk five years before it strikes - A. GREGORY, THE GUARDIAN, 2026

[54] Early Prediction of Heart Failure From Routine Cardiac CT Using Radiomic Phenotyping of Epicardial Fat - OIKONOMOU ET AL., JOURNAL OF THE AMERICAN COLLEGE OF CARDIOLOGY, 2026

*"**students learn more than twice as much** in less time with an AI tutor compared to an active learning classroom, while also being more engaged and motivated."* [55]

— **AI tutoring outperforms in-class active learning**, Nature, 2025

[55] AI tutoring outperforms in-class active learning: an RCT introducing a novel research-based design in an authentic educational setting - Kestin et al., Nature, Scientific Reports, 2025

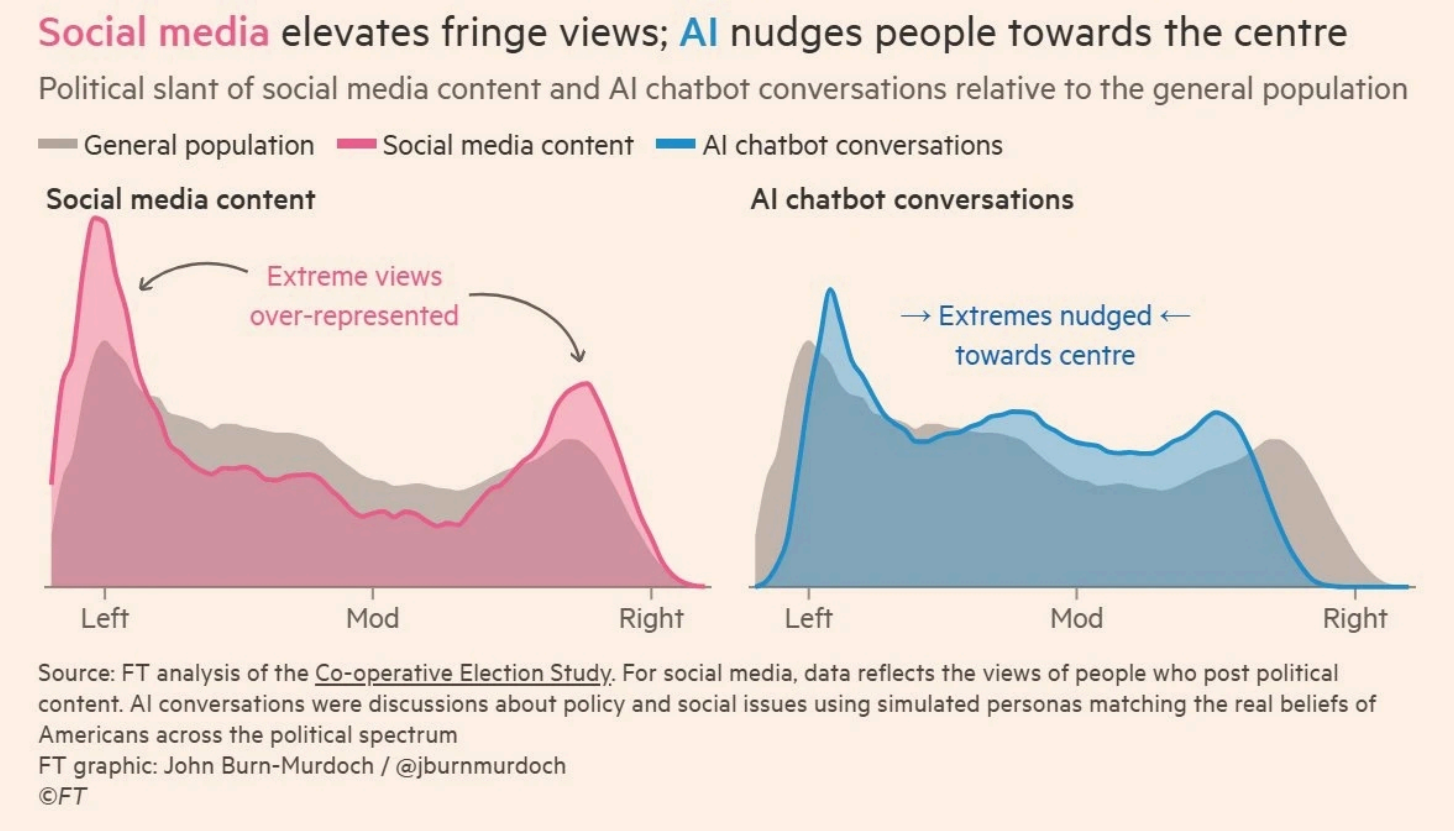
*"Our results show that personalized sequencing improves outcomes compared to fixed sequencing by 0.15 standard deviations on an in-person written exam completed without AI assistance, **which translates into as much as 6-9 months of additional schooling** according to some estimates. Notably, these gains were achieved without increasing instruction time or teacher workload. Furthermore, our mediation analysis suggests that these gains are mediated **by increased student engagement**"* [56]

— **Effective Personalized AI Tutors via LLM-Guided Reinforcement Learning**, Bastani et al., 2026

[56] Effective Personalized AI Tutors via LLM-Guided Reinforcement Learning - Bastani et al., Preprint, 2026

# Soften Polarization

- **AI may help to soften polarization.**




Social media is populist and polarising; AI may be the opposite - Financial Times, 2026

# 6. Negative Impacts

- **While data centers are not a concern on a global scale, they can severely impact local communities**. Their development can harm nearby communities through water use, air pollution, noise, competition for land, and energy cost increases. They can also reduce local tax revenues and typically do not provide other benefits such as high-paying jobs. [58, 59]

[58] From Energy Use to Air Quality, the Many Ways Data Centers Affect US Communities - Walker and Goldsmith, World Resources Institute, 2026

[59] What happens when data centers come to town? - Nguyen & Green, University of Michigan, 2025

"***The policy implication is therefore mixed. Data centers create economic activity, especially in directly related sectors and during construction***, *and they are associated with larger county-level income aggregates. They also raise electricity prices and are associated with higher house prices, which may benefit property owners while increasing costs for renters and prospective homebuyers.*" [60]

— **Data Centers and Local Economies in the Age of AI: A Shift-Share Approach**, National Bureau of Economic Research, 2026

[60] AI data centres can warm surrounding areas by up to 9.1°C - Alvarez et al., National Bureau of Economic Research, 2026

> *"... **residents report that they have had respiratory issues flare up** since xAI moved in.*
>
> *... Using satellite data, researchers at the University of Tennessee at Knoxville found that levels of nitrogen dioxide — which causes smog and is associated with asthma and other respiratory problems — near Colossus have been substantially elevated since its public announcement."* [61]
>
> — **Inside the Dirty, Dystopian World of AI Data Centers**, THE ATLANTIC, 2026

[61] Inside the Dirty, Dystopian World of AI Data Centers - M WONG, THE ATLANTIC, 2026

> *"A new data center being built with investments from Google will be partly powered by a natural gas project that emits the yearly emissions* ***equivalent of putting more than 970,000 additional gas-powered cars on the road****."* [62]
>
> — **A New Google-Funded Data Center Will Be Powered by a Massive Gas Plant**, Wired, 2026

[62] A New Google-Funded Data Center Will Be Powered by a Massive Gas Plant - Taft et al., Wired, 2026

- **Carbon emissions from AI are negligible compared to other sources**. Additionally, this technology has the potential to develop new materials [63], fund small nuclear reactors [64], and improve transportation planning and routing [65], which could contribute to reducing carbon emissions overall.

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[63] Millions of new materials discovered with deep learning - Merchant & Cubuk, DeepMind, 2023

[64] These nuclear companies are leading the race to build advanced small reactors in the U.S. - S. Kimball, CNBC, 2025

[65] GreenLight, Using Google AI to reduce traffic emissions - Google, 2025

> "***All data centers combined account for about 0.5% of global carbon emissions***. [66] *Of that, ~10% is AI-related, so AI is responsible for 0.05% of global carbon emissions. Cars and other transportation account for 25% of total emissions.*" [67]
>
> — **A. Holub**, 2026

> "*Robots are dramatically increasing efficiency in solar and wind projects, with systems capable of doubling installation speeds and improving precision.*" [68]
>
> — **Robots Are Quietly Building the Future of Renewable Energy**, OilPrice, 2026

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66 AI and climate change - International Energy Agency (IEA), 2025

67 A. Holub, 2026

68 Robots Are Quietly Building the Future of Renewable Energy - F. Bradstock, OilPrice, 2026

- **AI has demonstrated significant user deskilling**.

> *"AI use impairs conceptual understanding, code reading, and debugging abilities, [17% lower] without delivering significant efficiency gains on average."* [69]
>
> — **How AI Impacts Skill Formation**, Anthropic, 2026

> *"continuous exposure to AI for polyp detection [potentially cancerous] reduced the adenoma detection rate…with a 6.0% absolute difference, suggesting a detrimental effect on endoscopist capability."* [70]
>
> — **Rapid generative AI rollout in health care**, The Lancet, 2025

[69] How AI Impacts Skill Formation - Shen & Tamkin, arXiv, Anthropic, 2026

[70] Rapid generative AI rollout in health care - The Lancet Digital Health, 2025

> *"EEG analysis presented robust evidence that LLM, Search Engine and Brain-only groups had significantly different neural connectivity patterns, reflecting divergent cognitive strategies.*
>
> ***Brain connectivity systematically scaled down with the amount of external support**: the Brain-only group exhibited the strongest, widest-ranging networks, Search Engine group showed intermediate engagement, and LLM assistance elicited the weakest overall coupling. *" [71]*
>
> — **Your Brain on ChatGPT**, Kosmyna et al., 2025

* "The ChatGPT group showed notably less brain activity — it was reduced by up to 55%."

[71] Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task - Kosmyna et al., arXiv, 2025

AI chatbots could be making you stupider - BBC, 2026

*"Our findings demonstrate that people readily incorporate AI-generated outputs into their decision-making processes, **often with minimal friction or skepticism**.*

*These findings raise important questions about how decision-makers engage with AI under conditions of uncertainty or error. For example, in contexts such as financial advice, medical triage, or legal decision support, uncritical evaluation of System 3 **could result in significant harm and a lack of personal accountability for serious life outcomes.**"* [73]

— **Thinking-Fast, Slow, and Artificial: How AI is Reshaping Human Reasoning and the Rise of Cognitive Surrender**, University of Pennsylvania, 2026

[73] Thinking-Fast, Slow, and Artificial: How AI is Reshaping Human Reasoning and the Rise of Cognitive Surrender - Shaw & Nave, University of Pennsylvania, 2026

*"If you focus on just driving out results at the limit of whatever AI can do right now, you're only caring about the intercept, you know.* ***So I think it's basically a path to obsolescence through both the company and the people who are in it****. And so I'm really surprised how many executives of big companies are pushing this now ...*

*...* ***They're basically setting up their companies to be destroyed.****"* [74]

— **Jeremy Howard**,
The Dangerous Illusion of AI Coding?, 2026

[74] The Dangerous Illusion of AI Coding? - Jeremy Howard, 2026

> *So the default behavior is very similar to a self driving car. But there's this tipping point where at some point, you're not engaged anymore. You're not paying attention. And* ***you get this delegation of competence. And you get understanding debt.***
>
> *… But what happens is that the default attractor is for people to just go into this autopilot mode and they've got no idea what's happening and* ***it's actually making them dumber****".*

— **Jeremy Howard**,
The Dangerous Illusion of AI Coding?, 2026

*When many individuals consult the same centralized source of information, their beliefs and strategies tend to converge toward a common set of outputs, replacing previously heterogeneous private knowledge with shared knowledge derived from that source*

*... **This population-level loss of intellectual diversity could then negatively affect long-term problem solving ability of firms and undermine firms' competitive advantage*** [75]

— **Who Benefits from AI? Self-Selection, Skill Gap, and the Hidden Costs of AI Feedback**, Riedl & Bogert, 2026

[75] Who Benefits from AI? Self-Selection, SkillGap, and the Hidden Costs of AI Feedback - Riedl & Bogert, arXiv, 2026

*“ Students who learned without AI retained substantially more information after 45 days than those who used ChatGPT. The effect size corresponds to an ~11 percentage-point performance gap.*

*... **By providing immediate, comprehensive answers, the AI tool facilitated a form of cognitive offloading that eliminated the desirable difficulties needed for deep learning**. Skipping those effortful processes likely led to weaker memory encoding, as evidenced by the steeper forgetting curve in the AI-assisted group“* [76]

— **ChatGPT as a cognitive crutch: Evidence from a randomized controlled trial on knowledge retention**, Social Sciences & Humanities Open, 2026

[76] ChatGPT as a cognitive crutch: Evidence from a randomized controlled trial on knowledge retention - A. Barcaui, Social Sciences & Humanities Open, 2026

# Echo Chamber / Sycophancy 1/2

> *"We have provided both theoretical and empirical results showing that AI systems providing information that is informed by the user's hypotheses* ***result in increased confidence in those hypotheses while not bringing the user any closer to the truth****."* [77]
>
> — **A Rational Analysis of the Effects of Sycophantic AI**, Batista et al., 2026

[77] A Rational Analysis of the Effects of Sycophantic AI - Batista et al., arXiv, 2026

"***AI companies are making design choices that impact the psychology of billions of people worldwide***. *We found that brief interactions with* ***sycophantic AI chatbots lead to more extreme and certain beliefs — but greater enjoyment***. *Thus, AI companies face a tradeoff between creating engaging and enjoyable AI systems that foster 'echo chambers' or creating less engaging AI systems that may be healthier for users and public discourse.*" [78]

— **Sycophantic AI increases attitude extremity and overconfidence**, Rathje et al., 2026

[78] Sycophantic AI increases attitude extremity and overconfidence - Rathje et al., PsyArXiv, 2026

*"More than 5 million US youth (13.1%) have sought mental health advice from AI, **with rates reaching 22.2% among 18- to 21-year-olds**. A 2025 study of adults with mental health conditions who use large language models reported nearly half use them for support, including for anxiety, depression, and personal advice. Users seek emotional support, companionship, psychoeducation, and help processing difficult experiences, likely between sessions and **sometimes instead of clinical care altogether**."* [79]

— **Patients Use AI-Clinicians Should Ask How**, JAMA Psychiatry, 2026

[79] Patients Use AI-Clinicians Should Ask How - K. Saba & B. Weeks, JAMA Psychiatry, 2026

> *"Most interactions, however, come from non-experts using chatbots like search engines, including for everyday health and medical queries.*
>
> *… **Nearly half (49.6%) of responses were problematic**: 30% somewhat problematic and 19.6% highly problematic… **Chatbot outputs were consistently expressed with confidence and certainty** … Reference quality was poor… Chatbot hallucinations and fabricated citations precluded any chatbot from producing a fully accurate reference list."* [80]
>
> — **Generative artificial intelligence-driven chatbots and medical misinformation: an accuracy, referencing and readability audit**, BMJ Open, 2026

[80] Generative artificial intelligence-driven chatbots and medical misinformation: an accuracy, referencing and readability audit - Tiller et al., BMJ Open, 2026

- *“59% who use AI for health info are researching before doctor visits.*
- ***About 14 million adults report skipping a provider visit after using AI**.*
- *Only 4% who use AI for health info strongly trust its accuracy“.* [81]

— **Americans Turning to AI to Supplement Healthcare Visits**, Gallup, 2026

*“Consumers are adopting AI across healthcare touchpoints.*

***64% of respondents believe fluent AI users can perform at least one task as well as — or better than — a doctor**, including performing basic medical procedures 22%, and determining proper treatment or medication 21%.“* [82]

— **Edelman Trust Barometer**, 2026

[81] Americans Turning to AI to Supplement Healthcare Visits - Raynes & Maese, Gallup, 2026

[82] Edelman Trust Barometer - 16 Countries, 16,000+ Respondents, 2026

*"The condition doesn't appear in the standard medical literature — because it doesn't exist. It's the invention of a team led by Almira Osmanovic Thunström, a medical researcher at the University of Gothenburg, Sweden, who dreamt up the skin condition and then uploaded two fake studies about it to a preprint server in early 2024.*

*... Even more troublingly, other researchers say, **the fake papers were then cited in peer-reviewed literature**."* [83]

— **Scientists invented a fake disease. AI told people it was real**,
NATURE, FEATURE NEWS, 2026

[83] Scientists invented a fake disease. AI told people it was real - C. STOKEL-WALKER, NATURE, FEATURE NEWS, 2026

*"The apparent benefits of AI feedback are largely or entirely attributable to the same unobserved factors that drive AI adoption and engagement. Learning is concentrated among higher-skill, highly motivated individuals, due to their endogenous tendency to seek more AI feedback and use it more productively (after a failure). This individual-level pattern has important implications on the population level:* ***Instead of lifting lower-skilled individuals who have the most to gain, AI disproportionately complements higher-skilled individuals — amplifying the existing skill gap****."* [75]

— **Who Benefits from AI? Self-Selection, Skill Gap, and the Hidden Costs of AI Feedback**, Riedl & Bogert, 2026

- 41.3% of LinkedIn posts are generated by AI. [84]
- 71% of images shared on social media globally were AI-generated. [85]
- 35% of newly published websites were classified as AI-generated or AI-assisted. [14]

> *"an algorithmic shift from social-graph-based to interest-based recommendation, which is* ***remaking the active 'user' into a passive 'viewer'*** *the generative AI revolution, which is replacing user-generated content with synthetic media and* ***decoupling platforms from any dependence on human participation****"* [86]
>
> — **Towards a Post-Social Media Studies**, Törnberg & Rogers, 2026

[84] AI Vision - GPTZero, 2026

[85] AI Image Generator Market Statistics: An Analysis - ArtSmart, 2026

[86] Towards a Post-Social Media Studies - Törnberg & Rogers, SocArXiv, 2026

*"This is precisely why the most salient risk is not reducible to occasional inaccuracy or bias.* ***The risk is structural: correctness becomes decoupled from the processes of justification that normally sustain it****, and thus from the institutional and psychological practices through which* ***epistemic responsibility is enacted****."* [87]

— **Epistemological Fault Lines Between Human and Artificial Intelligence**, Quattrociocchi et al., 2026

[87] Epistemological Fault Lines Between Human and Artificial Intelligence - Quattrociocchi et al., PsyArXiv, 2026

*“LLMorphism: the biased belief that human cognition works like a large language model.*

*… **stronger LLMorphic beliefs will increase perceived replaceability of human workers, reduce perceived distinctiveness of human expertise, weaken attributions of agency and moral responsibility**, increase reliance on verbal fluency as a proxy for understanding, and reduce attention to embodied, affective, and contextual cues“* [88]

— **LLMorphism: When humans come to see themselves as language models**, V. Capraro, 2026

[88] LLMorphism: When humans come to see themselves as language models - V. Capraro, arXiv, 2026

The real danger to society is not that AI will actually become conscious, **but rather the risks that arise from *believing* it is**. We may grant legal rights or moral consideration to AI systems, or AI could be used to manipulate human users for commercial or **political gain**. [89, 90, 91, 92]

[89] Illusions of AI consciousness - Bengio & Elmoznino, Science, 2025
[90] 'Unethical' AI research on Reddit under fire - C. O'Grady, Science News, 2025
[91] AI deepfakes blur reality in 2026 US midterm campaigns - Ax & Coster, Reuters, 2026
[92] DHS is using Google and Adobe AI to make videos - J. O'Donnell, MIT Technology Review, 2026

*"Fusing LLM reasoning with multiagent architectures, these systems are capable of coordinating autonomously, infiltrating communities, and fabricating consensus efficiently.* ***By adaptively mimicking human social dynamics, they threaten democracy****."* [93]

— **How malicious AI swarms can threaten democracy**, SCIENCE, 2026

[93] How malicious AI swarms can threaten democracy - SCHROEDER ET AL., SCIENCE, 2026

*"prompting models in Chinese generates more positive responses about China's institutions and leaders than do the same queries in English. The combination of influence and persuasive potential across languages suggests the troubling conclusion that* ***states and powerful institutions have increased strategic incentives to leverage media control in the hopes of shaping LLM output****."* [94]

— **State media control influences large language models**, NATURE, 2026

[94] State media control influences large language models - WAIGHT ET AL., NATURE, 2026

> *"In debate pairs where AI and humans were not equally persuasive,* ***GPT-4 with personalization was more persuasive 64.4% of the time****...Our findings highlight the power of LLM-based persuasion and have implications for the governance and design of online platforms."* [95]
>
> — **On the conversational persuasiveness of GPT-4**, Nature, 2025

> *"I do not fear the rise of superintelligence.*
>
> *I do, however, fear the rise of billionaires, organizations, and world powers who seek to use computing to maximize their power, influence, and control."* [96]
>
> — **Grady Booch**, Creator of UML
> Author of the "Object-Oriented Analysis and Design" book

[95] On the conversational persuasiveness of GPT-4 - Salvi et al., Nature Human Behaviour, 2025

[96] Grady Booch - Bluesky, 2026